

SHAHZAIB SAQIB WARRAICH

warraich@usc.edu | (213) 681-6140 | [linkedin.com/in/shahzaib-saqib-warraich](https://www.linkedin.com/in/shahzaib-saqib-warraich) | <https://shahzaib-s-warraich.github.io>

EDUCATION

University of Southern California, Viterbi School of Engineering

Los Angeles, CA

M.S., Applied Data Science - CGPA: 4.00

Jan 2024 - Present

Coursework: Machine Learning, Deep Learning, Natural Language Processing, Database Management, Algorithm Design, Bias and Fairness in AI, Research Methods

EXPERIENCE

Graduate Research Assistant

Los Angeles, CA

Data, Interpretability, Language and Learning (DILL) lab

Jan 2024 - Present

Advisor: Dr. Swabha Swayamdipta

- Developing frameworks for GenAI (LLMs) to enhance explainability, interpretability, and evaluation of model reasoning.
- Designing models and systems for preference tuning of GenAI (LLMs) to improve reliability and trustworthiness.

Research Engineer

Redmond, WA

Retrocausal

Dec 2022 - Dec 2023

- Engineered precision assembly solutions with up to **98%** detection accuracy for industry leaders such as Schneider Electric, Carrier, and Honda using cutting-edge object detection and activity recognition algorithms.
- Led the development of an automated object detection model training pipeline, which employed the cut, paste, and mix augmentation technique to generate synthetic dataset alongside labeled data, thereby reducing manual data labeling time by **5x**.

AI Engineer

San Jose, CA

Adlytic AI

June 2021 - Dec 2022

- Led R&D on model optimization and pruning, leveraging knowledge distillation and Nvidia TensorRT to reduce edge inference latency by **3x** with less than **1.5%** accuracy loss.
- Developed and deployed a cloud-based recommender system pipeline, achieving up to **97%** accuracy on unseen data.

INDUSTRY PROJECTS

Person Re-identification System | PyTorch, Computer Vision, Nvidia TensorRT, FastAPI

April 2022 - Dec 2022

- Leveraged facial recognition, super resolution, object detection, and multi-object tracking algorithms to design the model inference API. Utilized RTSP to perform object tracking on frames from live streams.
- Optimized object detection models to achieve an accuracy of up to **93%** with an IoU of nearly **0.77**.

Mobile App Recommender System | PySpark, AWS, MongoDB, Redis, Docker

June 2021 - Mar 2022

- Designed end-to-end system architecture on AWS (SageMaker, EMR, EC2, S3, SNS).
- Optimized matrix factorization models to generate collaborative and content based recommendations using Apache Spark.
- Stored and served recommendations using MongoDB and AWS managed Redis databases.
- Packaged and deployed application using Docker containers.
- Boosted the product's installation and client conversion rate by nearly **200%**.

RESEARCH PROJECTS AND PUBLICATIONS

- Residual balanced attention network for real-time traffic Scene Semantic Segmentation (IJCECE 2023)
- Natural Language Generation Uncertainty and Variability Profiling (aiming for NeurIPS 2025)

LEADERSHIP

Viterbi Graduate Student Association (VGSA)

Senator

Aug 2024 - Present

- Representing a student body of nearly **2,000** students.
- Planning and organizing professional career development events.

SKILLS AND INTERESTS

Skills

Languages (Python, SQL), Deep Learning Frameworks (PyTorch, Hugging Face, vLLM), Machine Learning Frameworks (Scikit-Learn, OpenCV), Data Analysis Frameworks (Pandas, Dask, PySpark), Data Visualization (Matplotlib, Seaborn, Plotly), Cloud (AWS), Others (Docker, MongoDB, Git, LaTeX)

Interests

AI Ethics and Responsibility, Hackathons, User Experience (UX) Design